Microscopy

See & compare

1 Magnification & resolution



Blurred Separate points

Magnification: how much larger the image is.
Resolution: distinguish two nearby points as separate.

A blurry photo can be enlarged without adding detail. Bigger does not mean clearer.

2 Light & electron

Light microscopes show cells and visible structures.

Electron microscopes have much greater **magnification and resolving power** → reveal finer sub-cellular detail. Better resolution separates nearby details.

3 Scale & area

20 µm ÷ 2 µm = 10: **one order of magnitude.** Convert units before comparing.

Scale bar = actual distance; compare image length with it. Estimate relative sizes/areas when boundaries are unclear.

For similar shapes, length ×2 → **area ×4;** state this assumption.

Calculate size

4 Size formulae

Magnification = image ÷ actual size.

Actual = image ÷ magnification. Image = actual × magnification.

Use **the same units** before dividing. Magnification has no unit; sizes do.

5 Worked magnification

Image 30 mm; actual 60 µm. **30 × 1,000 = 30,000 µm.**

30,000 ÷ 60 = **×500.** Microscope total = eyepiece × objective: ×10 × ×40 = **×400.**

Drawing magnification depends on drawing size, not automatically the microscope setting.

6 Units & standard form

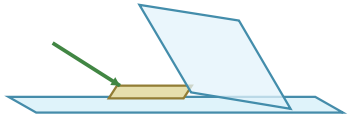
1 cm = 10 mm; 1 mm = 1,000 µm; 1 µm = 1,000 nm.

1 µm = 10^{-6} m; 1 nm = 10^{-9} m. Large → small units: multiply; reverse: divide.

$a \times 10^{n}$, **$1 \leq a < 10$.** Example: 0.000060 m = **6.0×10^{-5} m.**

Observe & record

7 Prepare the slide



Lower coverslip at an angle

Observe **plant and animal cells.** Thin onion epidermis + water; suitable stain (e.g. iodine) improves contrast.

Lower coverslip **at an angle** to reduce bubbles. Thin tissue lets light through and avoids overlap.

Use prepared animal cells or samples approved by your teacher. Handle glass, stains and samples safely.

8 Focus & draw

Lowest-power objective; **coarse then fine focus.** Centre cells; increase magnification; use fine focus.

Draw clear pencil outlines, no shading. Label identifiable structures; add **magnification scale or scale bar.** Interpret visible features, not memory alone.